A Coordinate-Safe Phase Portrait for the Bose–Josephson Dimer

HCS Research Program

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Abstract

We study the normalized two-mode Bose–Josephson dimer $H(z,\phi)=\Lambda z^2/2-\sqrt{1-z^2}\cos\phi$, $\Lambda\ge 0$. Passing to the Bloch sphere removes the apparent coordinate singularity at $z=\pm 1$. We classify $(0,0)$, $(0,\pi)$, and the $\Lambda>1$ symmetry-broken equilibria, including the $\Lambda=1$ pitchfork. Elimination of the phase gives an exact quartic in z; its roots yield complete-elliptic-K periods on crossing and self-trapped components. The $H=1$ sech homoclinic, its $\Lambda=2$ pole limit, and the reverse component criterion are stated without assigning a finite period to a separatrix. This is a source-local phase-portrait theorem: no arithmetic owner or target determinant is claimed.

1 Hamiltonian and Bloch regularization

Set

$$H(z,\phi)=\frac{\Lambda z^2}{2}-\sqrt{1-z^2}\cos\phi,\qquad \Lambda\ge 0,$$

with canonical equations

$$\dot z=-\sqrt{1-z^2}\sin\phi,\qquad \dot\phi=\Lambda z+\frac{z\cos\phi}{\sqrt{1-z^2}}.\tag{1}$$

The Bloch variables $x=\sqrt{1-z^2}\cos\phi$, $y=\sqrt{1-z^2}\sin\phi$ satisfy

$$\dot x=-\Lambda zy,\qquad \dot y=z(1+\Lambda x),\qquad \dot z=-y.\tag{2}$$

Thus the north and south poles have $\dot y=+1$ and -1, respectively; they are not equilibria and no value of the singular angle ϕ is assigned there.

Theorem 1 (Fixed points and pitchfork)**.** *The points $(0,0)$ and $(0,\pi)$ exist for every Λ. The first has energy -1, linearization $\left[\begin{smallmatrix}0 & -1\\ \Lambda+1 & 0\end{smallmatrix}\right]$, and frequency $\sqrt{\Lambda+1}$. The second has energy $+1$ and linearization $\left[\begin{smallmatrix}0 & 1\\ \Lambda-1 & 0\end{smallmatrix}\right]$: it is elliptic for $\Lambda<1$, parabolic at $\Lambda=1$, and hyperbolic for $\Lambda>1$. For $\Lambda>1$, two additional elliptic points are*

$$z=\pm\sqrt{1-\Lambda^{-2}},\quad \phi=\pi,\quad H_{\max}=\frac{\Lambda+\Lambda^{-1}}{2}.\tag{3}$$

In the canonical $(z,\phi-\pi)$ coordinates their matrix is $\left[\begin{smallmatrix}0 & 1/\Lambda\\ -\Lambda(\Lambda^2-1) & 0\end{smallmatrix}\right]$, so their small-amplitude period is $2\pi/\sqrt{\Lambda^2-1}$. The zero-phase crossing limit is $2\pi/\sqrt{\Lambda+1}$, and (3) coalesces with $(0,\pi)$ at the pitchfork $\Lambda=1$.

Proof. Differentiate (1) and solve $\dot z=\dot\phi=0$ away from the poles. The displayed matrices follow by first-order expansion. Equation (2) is obtained by the chain rule and extends the vector field over both poles. □

2 Energy quartic and regular periods

On $H=h$,

$$\dot z^2=(1-z^2)-\left(\frac{\Lambda z^2}{2}-h\right)^2=-\frac{\Lambda^2}{4}z^4+(\Lambda h-1)z^2+1-h^2.\tag{4}$$

For $\Lambda>0$, let

$$y_\pm=\frac{2(\Lambda h-1\pm\sqrt{\Lambda^2-2\Lambda h+1})}{\Lambda^2}.\tag{5}$$

Then $\dot z^2=(\Lambda^2/4)(y_+-z^2)(z^2-y_-)$. If $-1<h<1$, $y_-<0<y_+$, the level is one connected crossing component and

$$T_{\text{cross}}=\frac{8}{\Lambda\sqrt{y_+-y_-}}K\left(\sqrt{\frac{y_+}{y_+-y_-}}\right).\tag{6}$$

If $\Lambda > 1$ and $1 < h < H_{\max}$, there are two sign-preserving components and each has

$$T_{\text{self}} = \frac{4}{\Lambda\sqrt{y_+}} K\left(\sqrt{1 - \frac{y_-}{y_+}}\right). \tag{7}$$

Here K uses the modulus convention; the code passes its square to the numerical library. The limits near the two elliptic centers are exactly the frequencies stated above.

3 Separatrix and component criterion

For $\Lambda > 1, h = 1$, (4) has the homoclinic solution

$$z(t) = \pm\frac{2\sqrt{\Lambda - 1}}{\Lambda}\,\mathrm{sech}(\sqrt{\Lambda - 1}\,t). \tag{8}$$

The full critical level is connected through the saddle at $z = 0$, with two one-sided homoclinic branches. At its turning point the phase is π for $1 < \Lambda < 2$, the point is a Bloch pole at $\Lambda = 2$, and the phase is 0 for $\Lambda > 2$. Formula (8) has infinite period; it is not a regular periodic orbit.

For regular $\Lambda > 1$ levels, $1 < h < H_{\max}$ is exactly the self-trapped regime: the two connected components preserve the sign of z, and changing the initial sign selects the reflected component. For $-1 < h < 1$, the single component crosses $z = 0$, so both signs occur. At $h = 1$, sign change is only asymptotic. At $\Lambda = 1, h = 1$, (4) reduces to $-z^4/4$, so only the isolated degenerate point $z = 0$ remains. At $\Lambda = 0$, $H = -x$ gives rigid Bloch rotation of period 2π on regular circles.

References

[1] A. Smerzi, S. Fantoni, S. Giovanazzi and S. Shenoy, Quantum coherent atomic tunneling between two trapped Bose–Einstein condensates, *Phys. Rev. Lett.* 79 (1997), 4950–4953, DOI: 10.1103/PhysRevLett.79.4950, `https://doi.org/10.1103/PhysRevLett.79.4950`.

[2] S. Raghavan, A. Smerzi, S. Fantoni and S. Shenoy, Coherent oscillations between two weakly coupled Bose–Einstein condensates, *Phys. Rev. A* 59 (1999), 620–633, DOI: 10.1103/PhysRevA.59.620, `https://doi.org/10.1103/PhysRevA.59.620`.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic, target-zero, determinant-matching, or Hilbert–Pólya claim. **Data and code.** All rows and audits accompany HCS-C243. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; the artifact chain checks displayed formulas and metadata.